Find the missing bridge

Identify what must carry over when someone extends a result to a new setting.

Suggested teaching time: 20 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None

Students will be able to

- Separate an observed result from a prediction about a new setting.
- State a relevant assumption connecting the two.
- Describe a possible difference that would weaken that connection.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; no student devices required.
- Keep the instructor guide and model answers separate from student materials.
- Read the three scenarios and their model answers.
- Print the student handout or display one activity at a time.
- Explain that all scenarios are invented and students should use only the supplied facts.

Teaching sequence

4 min	**Notice the untested step.** Learners complete the bus-route warm-up. Name the gap between a result on one route and a prediction for another.
7 min	**Find the checkout assumption.** Pairs separate the observed result, prediction, and assumption in the main activity. Ask for one relevant possible weekday difference.
4 min	**Test the bridge.** Discuss the supplied prompts and reveal the main answer. Distinguish a possible weakening condition from an established fact.
5 min	**Check independent understanding.** Learners complete the practice-session exit ticket alone. Assess whether the assumption connects the specific settings.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / A faster bus route

Student task and instructor answer

Explain these words as needed

Assumption: An idea the reasoning relies on without stating or establishing it.

Prediction: A claim about what will happen or what an untested case will show.

Relevant difference: A difference that could affect whether a result carries over.

Scenario

Invented scenario: An express service reduces journeys on Route A by ten minutes. A planner says, 'The same express arrangement will save ten minutes on Route B.' Route B has not been tested.

Student output: Write the observed result, one assumption, and one possible difference.

1. State what was observed on Route A.

2. Name a feature that would need to be similar enough for the same time saving on Route B.

3. Describe a possible Route B difference that could reduce the saving.

Student success check: Is your difference relevant to journey time, and clearly a possibility rather than a supplied fact?

Model answer

Route A journeys became ten minutes shorter. The planner assumes relevant conditions, such as the stops that can be skipped, are similar enough on Route B. If Route B has fewer skippable stops, the saving could be smaller; that difference is only a possibility here.

Why this meets the task: The answer connects the observed result to the untested route without inventing information about Route B.

Main activity / Will the checkout result carry over?

Student task and instructor answer

Scenario

Invented scenario: A store tests a checkout arrangement on Saturday and Sunday. Average waits fall from six minutes to four. The manager concludes, 'This arrangement will also reduce waits on weekdays.' No weekday test has been done.

Student output: Write the observed result, prediction, assumption, and one possible weakening difference.

1. Copy the observed reason and the prediction separately.
2. Complete: 'The manager assumes that weekday conditions are similar enough in ___ for the weekend result to carry over.' Name a relevant feature such as customer volume or staffing.
3. Describe one possible weekday difference that would weaken that bridge. You are proposing a possibility, not claiming it happened.

Student success check: Does your assumption connect this weekend result to weekdays, rather than merely say that faster checkout is desirable?

Model answer

Reason: weekend waits fell from six to four minutes. Prediction: weekday waits will fall too. The manager assumes weekday customer volume and staffing are similar enough for the arrangement to help. If weekdays have much less staff available, the arrangement might not work as it did in the trial.

Why this meets the task: It identifies the needed transfer from one setting to another and tests that assumption with a relevant possible difference.

Exit ticket / A practice session for another class

Student task and instructor answer

Scenario

Invented scenario: A teacher gives her morning class ten minutes of vocabulary practice each day. Its average score rises from 60 to 75 on a weekly vocabulary quiz. She predicts the same routine will raise the evening class's score by 15 points. The evening class has not tried it.

Student output: Write the observed result and prediction, then an assumption and a possible weakening difference.

1. Separate the morning result from the evening prediction.

2. Name one learning condition that would need to be similar enough for the prediction.

3. Describe a possible difference that could make the evening class improve less.

Student success check: Does your assumption concern learning conditions, rather than just the desirability of higher scores?

Model answer

Observed: the morning class average rose from 60 to 75. Prediction: the evening class will gain 15 points. The teacher assumes relevant conditions, such as starting vocabulary knowledge, are similar enough. If evening learners already know most of the practiced words, the same routine might produce a smaller gain. That condition is not established here.

Why this meets the task: The answer makes the transfer assumption explicit and marks the weakening difference as hypothetical. It does not claim that the before-and-after result proves practice caused the rise.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- The observed result and untested prediction are kept separate.

- The assumption names a condition relevant to carrying the result over.

- The possible difference is explained and not presented as an established fact.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Must the two settings be identical in every way?

Listen for: No. The assumption concerns similarities that matter for this result. Wall color need not matter to checkout waiting time; staffing might.

Does naming a possible difference refute the prediction?

Listen for: No. A possibility identifies something to investigate. It is not evidence that the difference actually exists or that the prediction is false.

Common confusions

An assumption is always false. It may be reasonable or true, but the argument still needs it. Ask what evidence would support it.

Any difference defeats a comparison. Explain how the difference could affect this outcome. A difference with no such connection is insufficient.

The observed change establishes its cause. These before-and-after reports leave other explanations open. Finding a transfer assumption does not settle the causal question.

Adapt the lesson

More support: Use the frame: "The result was observed in ___. The prediction concerns ___. The reason carries over only if ___ is similar enough." Accept one relevant condition.

More challenge: Ask what evidence would check the weekday staffing assumption. Expected answer: compare weekend and weekday staffing and customer volume, then test the arrangement on weekdays; these are proposed checks, not completed studies.

Next practice

https://xixteen.com/learners/?practice=hidden_assumptions#workbench